Qihan (Blake) Wang

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Education

University of Illinois Urbana-Champaign Urbana-Champaign, IL

B.S. in Electrical Engineering *Expected May 2029*

- **Relevant Coursework:** Differential Equations; Calc III; Linear Algebra; Discrete Structures; Intro to Electronics

Experience

CUDA-Accelerated Nuclear Pasta Research Oct. 2026–Present

UIUC | Research with Dr. Soham Pal | CUDA C++, Linux, ICRN

- Implemented an FP64 CUDA kernel for all-pairs nuclear and screened Coulomb force calculations using a structure-of-arrays layout, periodic boundaries, and shifted-force cutoffs.
- Built CPU/GPU numerical validation and repeatability tests; implemented CUDA-event benchmarks with persistent device buffers and explicit host–device memory transfers.

Illini Solar Car Sept. 2026–Present

UIUC | Sensor Integration, Zephyr RTOS, KiCad, MCUXpresso, PCB Design

- Evaluating weather-resistant blind-spot sensors for solar car Delos; targeting side/rear obstacle detection powered by the existing 12 V distribution system.
- Configured a preliminary Zephyr RTOS environment for planned hardware-in-the-loop (HIL) testing ahead of final electrical-component selection.
- Designed an LED/resistor PCB in KiCad and wrote MCU firmware for potentiometer-controlled LED brightness using MCUXpresso IDE.

Computer Science/Cybersecurity Research Intern Summer 2025

University of Houston | Research with Dr. Weidong Shi | Python, pandas, Google Colab

- Developed Python scripts to evaluate large language model reasoning accuracy and cross-domain knowledge transfer across biology, mathematics, and law.

Projects

Pipelined INT8 FPGA Accelerator | *SystemVerilog, cocotb, Yosys/nextpnr, Python* 2026

- Implemented an 8-lane signed INT8 dot-product engine with exact 19-bit accumulation and ready/valid backpressure; achieved a no-stall initiation interval of one cycle with a four-stage pipeline.
- Verified 13,846 results per architecture against a Python reference under randomized stalls; Artix-7 synthesis mapped the pipeline to 8 DSP48E1 blocks versus 1 for a serial baseline.
- Completed iCE40UP5K place-and-route at a 12 MHz clock constraint; generated a bitstream and passed post-synthesis functional simulation.

EEG Motor-Imagery Classification and Signal Processing | *Python, MNE, SciPy, scikit-learn, PyWavelets* 2026

- Built an MNE preprocessing pipeline for EDF recordings from the PhysioNet EEG Motor Movement/Imagery Dataset, implementing bandpass filtering and epoch segmentation.
- Implemented CSP/FBCSP spatial feature extraction and wavelet-energy features; evaluated shrinkage LDA classification using cross-validation.
- Evaluated cross-subject transfer with few-shot calibration and assessed sensitivity to EEG artifacts.

Chronodox Strategic Simulation Platform | *React, TypeScript, Zustand, Vite, Node.js* 2026

- Developed a TypeScript strategy engine with procedural map generation and a browser-based command interface for gameplay and diagnostics.
- Built a custom snapshot serializer connecting client-side Zustand state to an in-memory Node.js room server; synchronized clients through HTTP polling and state pushes, with room revision checks rejecting stale updates.
- Developed browser-based self-play pipelines and trained action-ranking weights for a gameplay agent using 200,000+ completed self-play samples.

Activities & Leadership

CyberPatriot | *5th Place, NJROTC Platinum Semifinals* 2024–2025

- Hardened Linux systems in virtualized environments using Bash, system administration, and cybersecurity tools.

National Program Director, Steel City Codes 2025–2026

- Supported computer science education through curriculum development, outreach, and program coordination; created digital outreach content generating approximately 50,000 views.

Technical Skills

Languages: C/C++, Python, SystemVerilog, Verilog, Bash, TypeScript, Java
Hardware & Embedded: RTL design, FPGA synthesis/place-and-route, cocotb, Icarus Verilog, Yosys, nextpnr, KiCad, CUDA, Zephyr RTOS, MCUXpresso IDE
Tools & Systems: Git, Vite, npm, Jupyter Notebook, Linux
Scientific Libraries: NumPy, pandas, Matplotlib, MNE, SciPy, scikit-learn, PyWavelets
Web Technologies: React, Zustand, Canvas2D, Node.js, HTML, CSS